DASH Azure Kubernetes Service

Prerequisites

1. Azure CLI: Install and configure the Azure CLI on your local machine to run the commands below.
    - **Install Azure CLI**

2. Kubernetes CLI tool to interact with your cluster.
    - **Install Kubernetes CLI**

Input Guide

This outlines the input parameters available during **Azure Kubernetes Service (AKS) provisioning via the DASH Portal**.Each row describes a specific parameter, including its meaning, possible values, recommended values, and any valid combinations.

Input Parameter	Description	Values Provided	Value Guidance	Valid Combinations
cluster_name	Name of the AKS cluster.	User-defined string e,g. akscluster	Must be unique within the subscription.	N/A
region	Azure region where the AKS cluster will be deployed.	East US, Central US, West Europe, Central India, UK South, Canada Central, East Asia, etc.	Choose the one closest to your workload for better latency.	N/A

kubernetes_version	Kubernetes version for the cluster.	Versions available: 1.31.7, 1.31.6, 1.31.5	Recommended to use the Latest version unless a specific version is required.	Must be a supported AKS version.
node_count	Number of nodes in the AKS cluster.	Numeric value 1,2,3,4,5,6	Used for fixed-size node pools.	Mutually exclusive with autoscaling settings.
minimum_nodes	Minimum number of nodes when autoscaling is enabled.	Numeric value 0,1,2,3	Recommended starting value: 1 or 2.	Must be ≤ maximum_nodes.
maximum_nodes	The maximum number of nodes when autoscaling is enabled.	Numeric value 1,2,3,4,5,6	Recommended: depends on workload scaling.	Must be ≥ minimum_nodes.
vm_size	Size of the Azure VM for worker nodes.	Examples: Standard_DS1_v2 Standard_DS2_v3 Standard_D2as_v4,Standard_B2s	Choose based on workload performance needs.	N/A
disk_size_gb	OS disk size in GB for AKS nodes.	Disk size available 30,40,50,60..100	Must be > 30 GB	N/A
owner	The team or individual responsible for provisioning, managing, and maintaining the resource.	Specify the team name or individual responsible for maintaining the cluster.	Provide correct team/individual name.	N/A

application	Associated service or application name for the resource.	Enter the associated service name linked to the cluster.	Provide a service name linked to the cluster.	N/A
project	Project or initiative name linked to the resource.	Indicate the project or initiative name related to the resource.	Provide the project name related to the cluster.	N/A
deployment	The target environment (e.g., development, testing, prod) where the cluster will be provisioned.	Define the deployment environment name (e.g., dev, test, prod)	Choose the environment where the cluster has to be deployed.	N/A
Services expire after 30 days. Create this service now, or schedule for a later date?	Cluster will be deleted 30 days after creation, or it can be scheduled to be created at a later date.	Create Now (or) Schedule for Later	Choose based on cluster requirements.	N/A

Sample Input Selection:

KUBERNETES VERSION	1.31.7
NODE COUNT	2
MINIMUM NODES	1
MAXIMUM NODES	3
CLUSTER NAME	akscluster
DISK SIZE	40
VM SIZE	Standard_DS2_v2
REGION	East US
OWNER	dashuser
APPLICATION	dashapp
PROJECT	dash
DEPLOYMENT	dev

Services expire after 30 days. Create this service now, or schedule for a later date?

(●) Create Now ( ) Schedule for Later

ADD SERVICE CANCEL

Output Guide:

After provisioning the **Azure Kubernetes Service (AKS) cluster**, the following output values are available.These outputs are essential for accessing the Kubernetes API server, configuring authentication, and managing workloads within the cluster.

Output	Description	Example Value
resource_group_name	Name of the resource group containing the AKS cluster.	dash-aks-rg
aks_cluster_name	Name of the provisioned AKS cluster.	dash-aks-cluster1
kube_config	Raw Kubernetes config file used to authenticate with the cluster.	Sensitive value (YAML content)
cluster_endpoint	API server endpoint (control plane URL) for the cluster.	https://dash-aks-cluster1-12345.hcp.eastus.azmk8s.io:443
cluster_credentials	Command to fetch cluster admin credentials.	az aks get-credentials --resource-group dash-aks-rg --name dash-aks-cluster1
cluster_access	Step-by-step instructions for logging in and accessing the cluster.	Multi-line Azure CLI steps
Delete_resource_group	Command to delete the AKS resource group when no longer needed.	az group delete --name dash-aks-rg --yes --no-wait

What You Can Do with These Outputs

This document outlines the output values generated after **Azure Kubernetes Service (AKS) provisioning** through the DASH Portal.These values can be used to securely access, manage, and interact with your Kubernetes cluster.

Access the AKS Cluster

1. Copy the kubeconfig output into a file

Copy the value shown under kube_config output.

Save it into a new file called kubeconfig-aks.yaml.(e.g)

2. Example:

Linux/macOS:

```
vi kubeconfig-aks.yaml

# paste the content, then save and exit
```

Windows (PowerShell):

```
notepad.exe kubeconfig-aks.yaml

# paste the content, then save
```

3. Set the kubeconfig environment variable

Linux/macOS:

```
export KUBECONFIG=$PWD/kubeconfig-aks.yaml

kubectl get nodes
```

Windows PowerShell:

```
$env:KUBECONFIG = "$PWD\kubeconfig-aks.yaml"

kubectl get nodes
```

Windows CMD:

```
set KUBECONFIG=%cd%\kubeconfig-aks.yaml

kubectl get nodes
```

4. (Optional) Merge with existing kubeconfig
If you already have other cluster configs:

Linux/macOS:

```
export KUBECONFIG=$PWD/kubeconfig-aks.yaml:~/.kube/config

kubectl config view --merge --flatten > ~/.kube/config
```

Windows PowerShell:

```
$env:KUBECONFIG = "$PWD\kubeconfig-aks.yaml;$HOME\.kube\config"

kubectl config view --merge --flatten | Out-File -Encoding utf8
$HOME\.kube\config
```

Switch between contexts if multiple clusters exist

```
kubectl config get-contexts

kubectl config use-context <context-name>
```

Recommendation: Instruct users to always copy the `kube_config` output into a dedicated file before setting `KUBECONFIG`.

Delete the Cluster

Once your work is complete, use the Delete_resource_group output to clean up all resources:

```
az group delete --name <resource_group_name> --yes --no-wait
```